

Sai Smrithi - Intro

I have been a Data Analyst for over 4 years.

Have hands on experience in **Looker, Tableau, Snowflake, Python, SQL**.

Worked with Monetate most recently :

- Closely collaborated with the Customer Success and the Product Teams to develop **Looker dashboards** related to Customer Funnel Metrics, RoI, Platform Usage and other ad hoc reports.
- **Optimized reports** and improved backend data pipelines using Python and modified load process to load Snowflake tables used by these reports.
- **Documented** the business and data flows and helped the customer success team in creating the self-serve dashboards for their adhoc presentations / discussions with their clients.

Wood U Like:

- Analyzed sales and inventory data.
- Developed **Tableau dashboards**
- provided accurate prediction models for their inventory and personalizations that helped reducing their shipping times from 7 days to 4 days.

Jetzy:

Apart from being a **Tableau developer** to prepare and present reports regarding their app usage and various key metrics, I was working on managing the dev team and establishing **project management** frameworks using tools such as JIRA / github.

Platform Usage Score: Analyzing Feature Adoption Trends

- Sai Smrithi Sai Prabhakar



Task

Present about a KPI that you created from scratch. How you validated the data source, and convinced leadership to adopt that KPI.

Presentation framework

- Overview of Monetate and high level details about its products, features and add-ons.
- Need for a new metric - the underlying business use case
- Creation of platform usage score metric
- Metric validation and rollout.
- The positive impacts of the metric and the associated changes to the business flow.

Monetate's Core Feature Ecosystem

Dynamic Testing

Shows best performing variant to the client, dynamically before the test completes to improve revenue.

Automated Personalization

Recommends content to the users based on their past behavioral patterns.

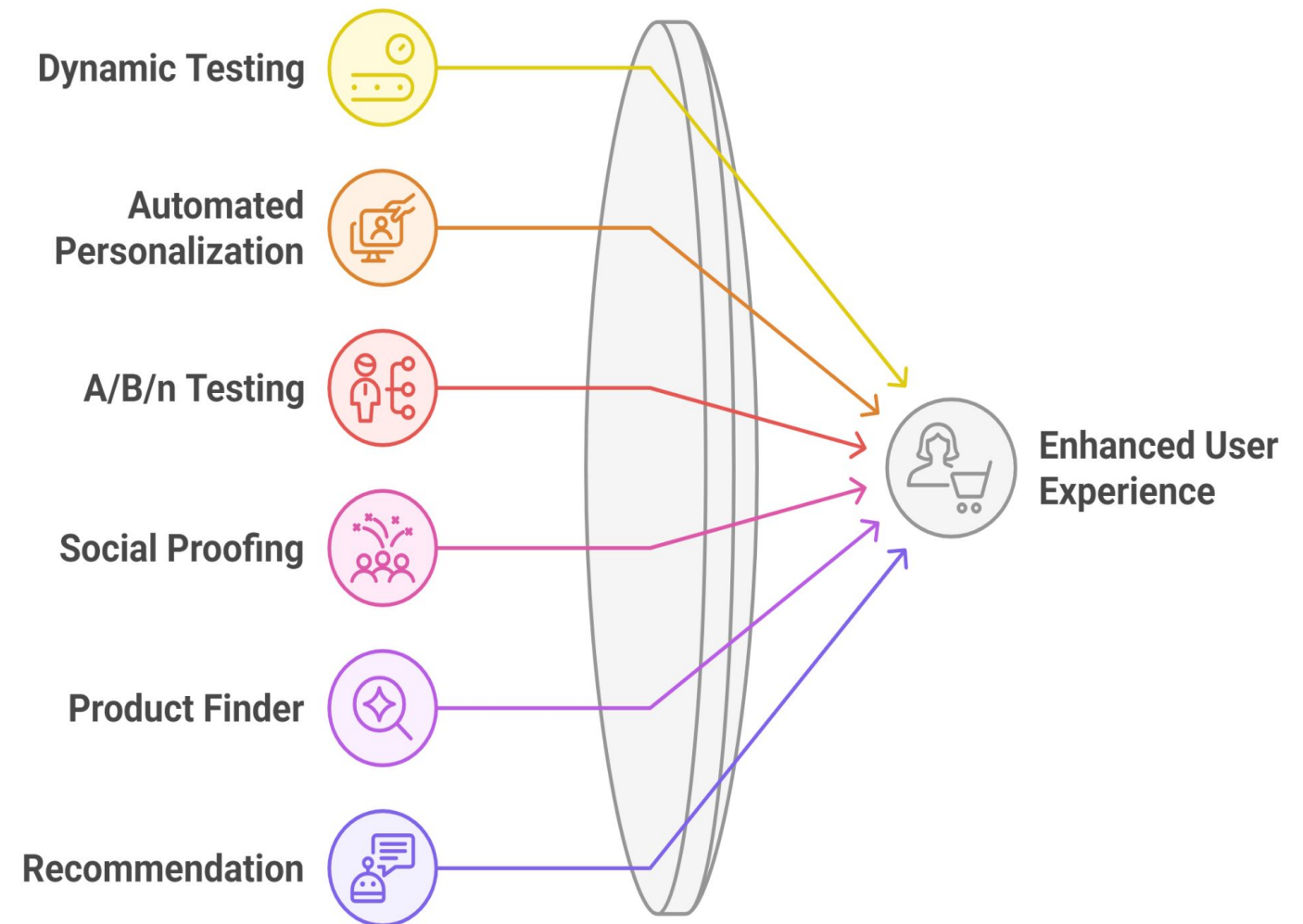
A/B/n Testing

Implements multiple comparisons (A, B, C, etc.) to determine best-performing options using statistically valid sample sizes

Social Proofing

Enhances shopping urgency with dynamic annotations like 'Only 2 left' or '100 people have bought this in the last 24 hours', etc. based on the real-time demand

Monetate Core Features



Monetate's Core Feature Ecosystem

Product Finder

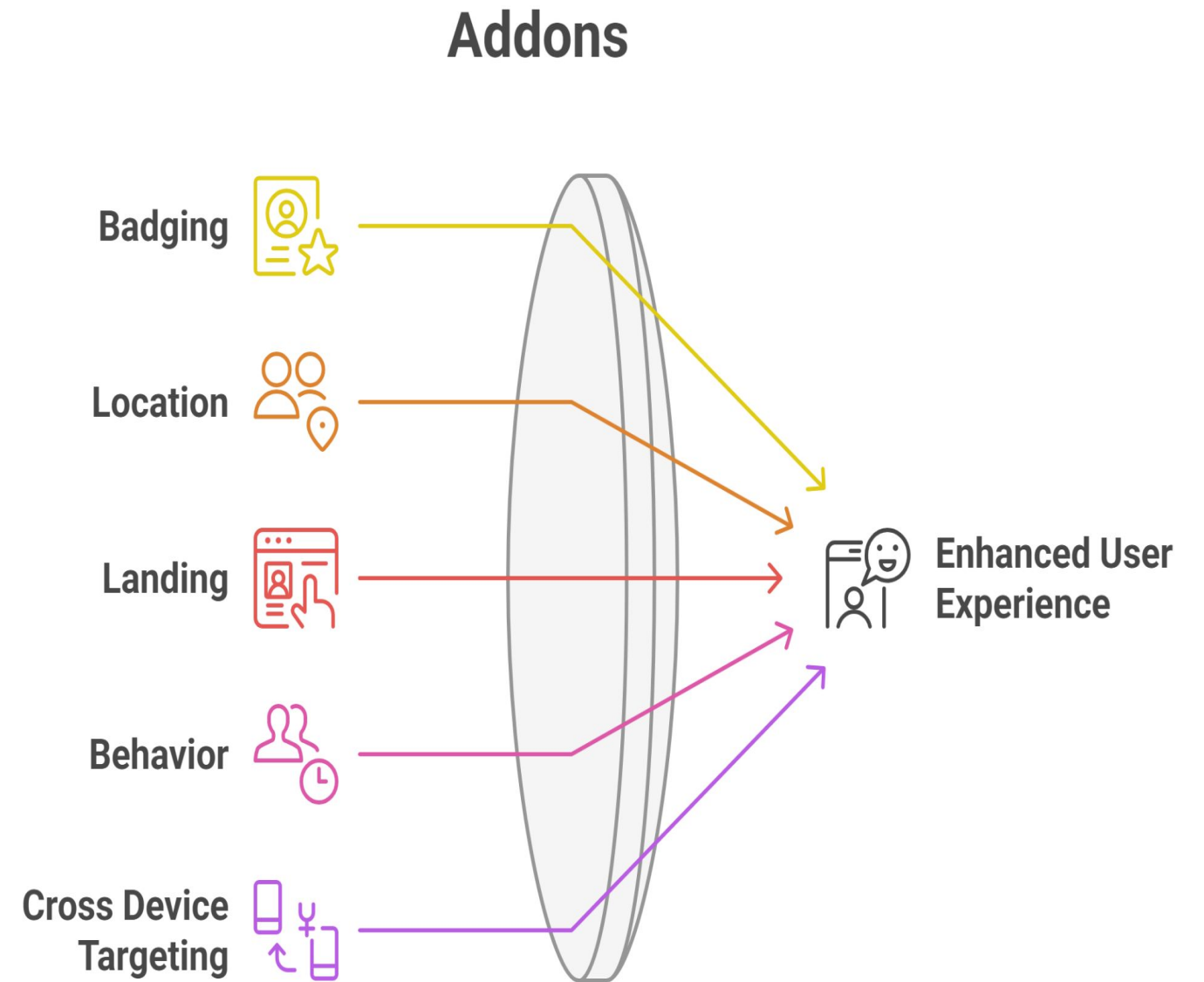
Uses customer-entered information via questionnaires to recommend tailored product options

Recommendation

Suggests additional products based on customer behavior and similar purchase patterns

Addons

Including badging, location, landing, behavior, and cross device targeting capabilities



The Ask

The request was from customer success team

- Monetate's **active client base was increasing** but there was not much improvement in the **retention rate**.
- Customer success team wanted to improve the retention rate.

Charting the way forward

1

Understanding the ask

Defining success and ideation
Phase : I discussed with the customer success team about

- What is required?
- What data is currently available?
- Different dashboards that they were currently using.
- Identifying the goal of this exercise

Understand the ask and identify additional details that could be used to improve retention rate.

2

Identify available information

- Backend data with the usage count of clients who used a particular feature
- Campaign backend table that records all the actions a client performed while using Monetate APIs / UIs
- Read through confluence documents to learn data architecture.

3

What is currently missing?

- When did the client activate a feature?
- When was the feature last used by the particular client?
- Ways to monitor the recent usage trends
- Unified metrics to help customer success team to effectively engage with their clients.

4

Bridging the gap

- Circled back with the team to discuss about the existing reports and ways we could show the data.
- Discuss about a unified platform usage score and a weighted decay score to efficiently portray the usage patterns of the clients.
- About time that could take to develop and test the metric.

Need for a new usage metric

No Last Usage Visibility

Dashboards lacked info on most recent feature use dates.

Cumbersome Analysis

Manual verification of feature usage was time-consuming for teams.

Unifying the View

Customer success needed consolidated metrics to improve client engagement and upsell products.

Developing the Platform Usage Score

Calculate Decay Score

Created a metric that evaluates retailer interaction with features over the past six months, with more recent activity contributing higher value

Apply Weights for features

Implemented 2x weights for foundational features (Dynamic Testing, Automated Personalization) versus standard weights for supporting flags

Compute Final Score

Combined decay scores and weights to produce a comprehensive Platform Usage Score

Calculating the Decay Score

Decay scores measure recent interaction within a 6-month window, emphasizing recency.

1.00

This Month

Indicates maximum engagement

0.80

1 Month Ago

0.66

2 Months Ago

0.00

6 Months Ago

Shows disengagement

Calculate the date difference between the current `month_start` and `max(last_usage_time)` observing a rolling window for the past 6 months

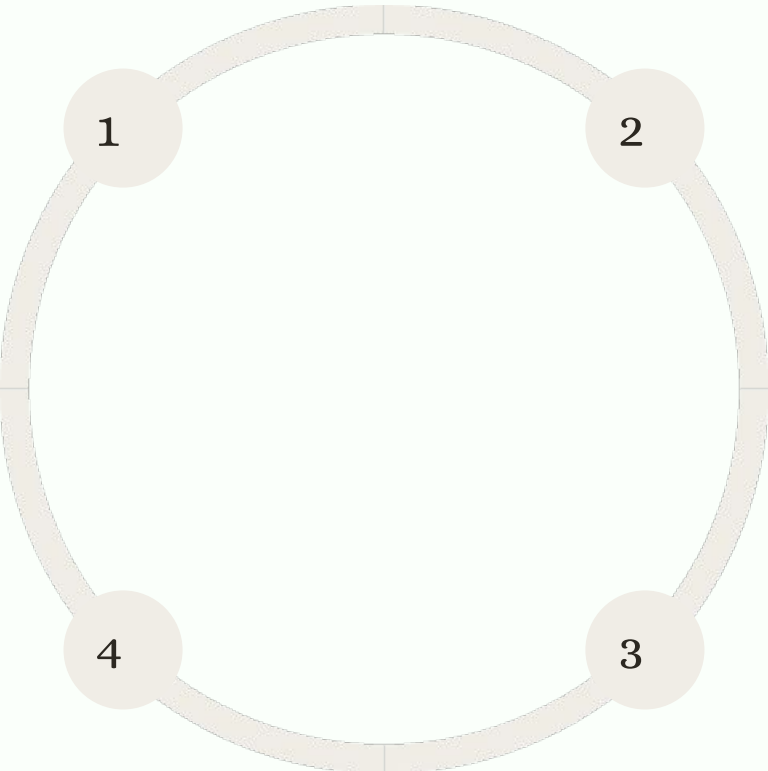
Weighted calculation strategy

Foundational Features

Weighted 2x because they directly impact conversion rates.

Outcome

Weighted scoring aligns metrics with business impact for adoption strategies and retention.



Addons

Standard weighting reflects enhancement, not core engagement.

Strategic Value

Major features must be active before add-ons can function.

Resolving data ambiguities



Gathered Raw Data

- Gathered raw data from MySQL tables from the campaign backend tables
- The field last usage time for a particular client gave details about when the client had switched on a flag related to these features in the backend.



Data Cleaning

Dealt with the NULL values in the last usage time.

1. The client was not aware of additional features that they could use.
2. The client was not allowed to use the feature because of his subscription package.

Resolving data ambiguities

Followed a structured approach



Engage Data Engineering Team

Collaborated with the team to clarify the meaning of null values in usage data and gain insights into activation status information.



Incorporate Client Tier Information

Leveraged audit database to understand subscription levels and associated feature access rights for different client tiers.



Join Multiple Data Sources

Combined client information, activation times, tier-to-feature mapping, and client-tier associations to create a comprehensive view.

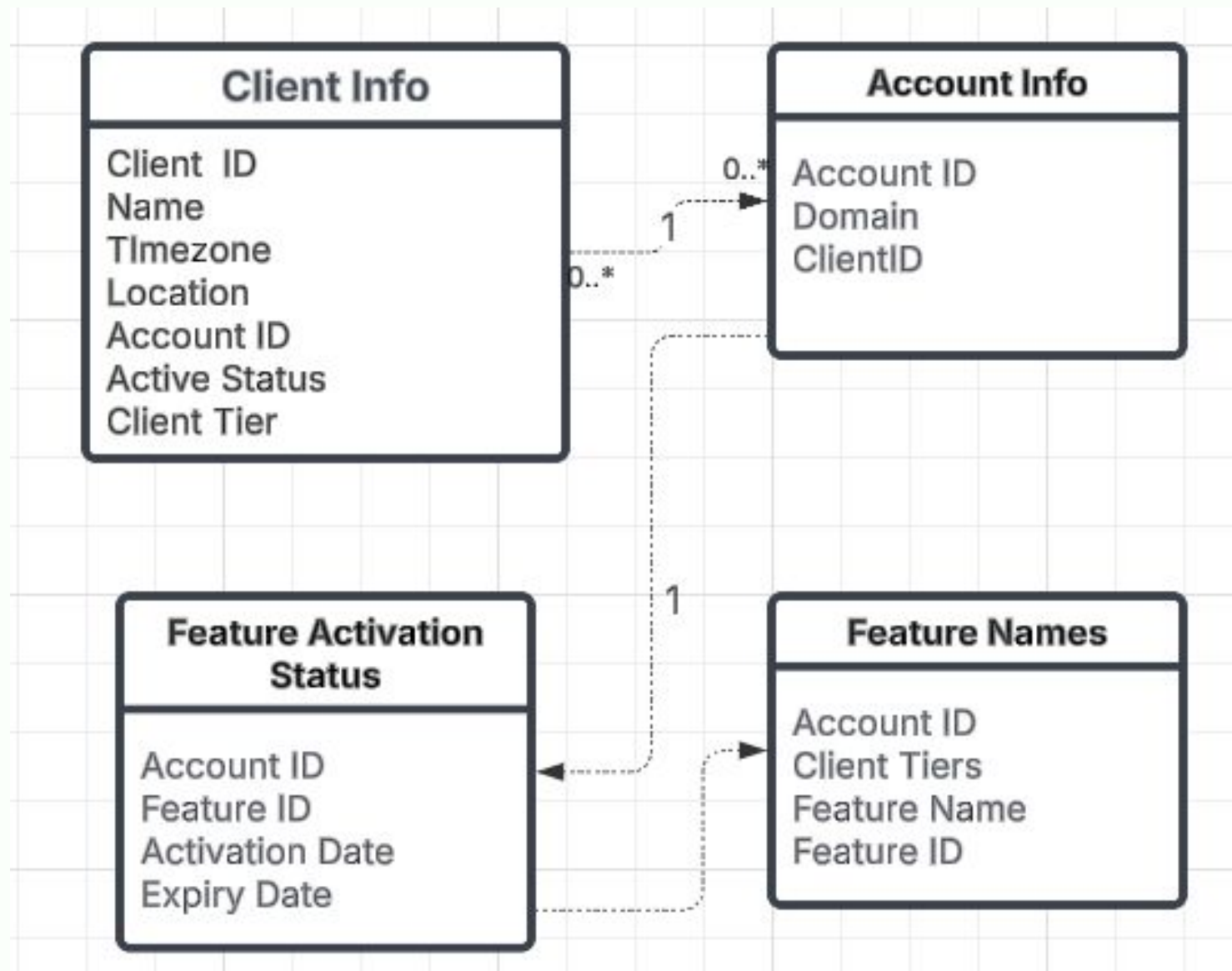


Determine Accurate Activation Dates

Formulated queries that precisely identified when each client gained access to specific features.

Deriving activation information

The Audit database had multiple tables which helped to derive the activation information of the clients.



The clean dataset

- By combining these data points, we were able to formulate a query that accurately determined the activation date for each feature per client.
- Once the data cleaning was complete and the relationships established, the refined data set was built by joining the usage table with the audit-derived activation data.

Column Name	Datatype
Account ID	Integer
Domain	String
Feature Name	String
Month Start	Timestamp
Last Usage Time	Timestamp

Applying Feature Weights in SQL

```
weighted_decay AS (  
    SELECT account_id,  
           month_start,  
           domain,  
           feature_name,  
           max_last_usage_time as last_used_on,  
           month_diff,  
           decay_score,  
           (CASE WHEN feature_name IN  
                ('Dynamic Testing', 'Auto Personalization', 'A/B/n Testing',  
                 'Social Proofing', 'Product Finder', 'Recommendation')  
                THEN 2*(decay_score) ELSE decay_score END) as weighted_decay  
    FROM decay_score
```

Final Usage Score and Aggregation

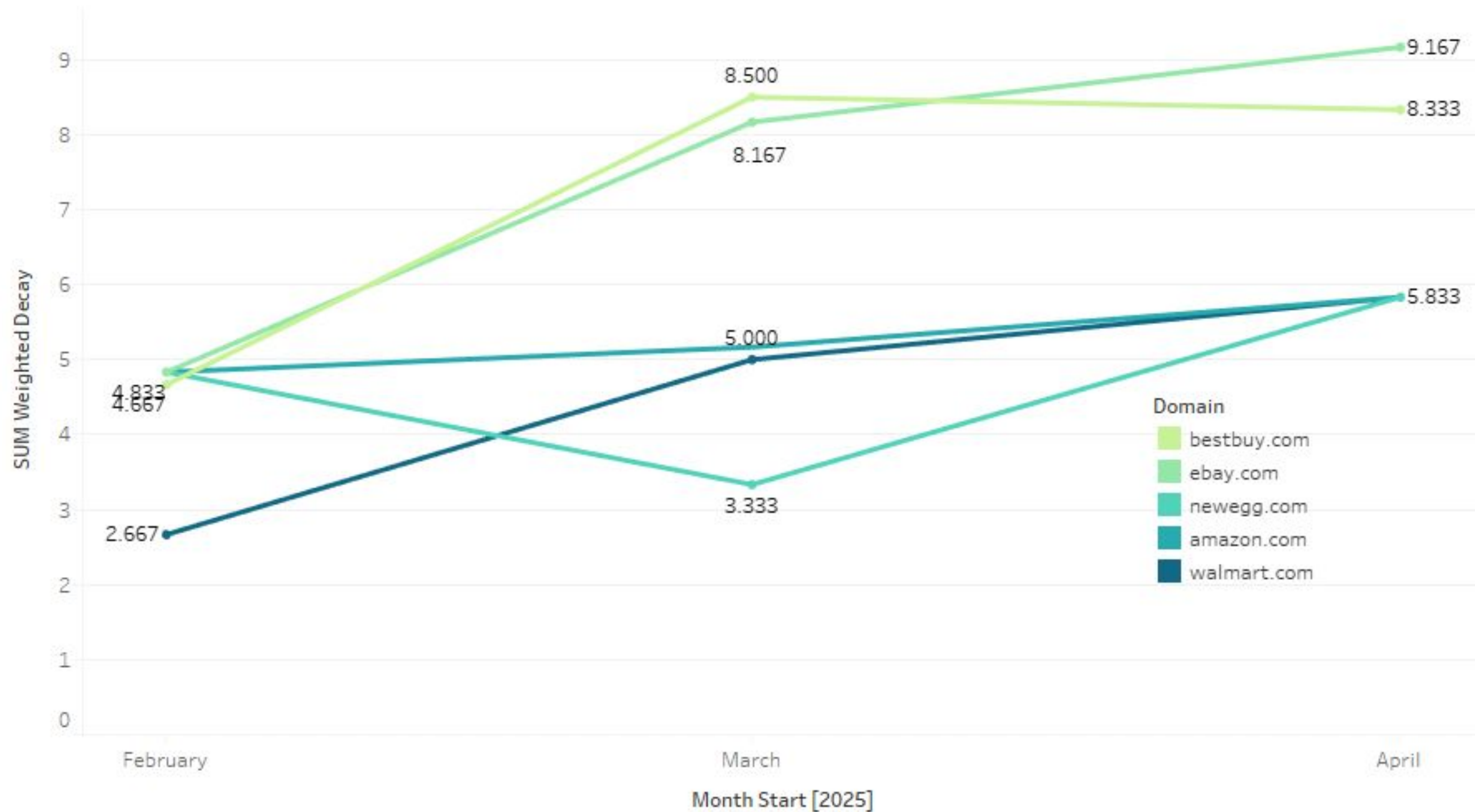
```
SELECT
    account_id,
    domain,
    month_start,
    feature_name,
    last_used_on,
    month_diff,
    decay_score,
    weighted_decay,
    SUM(weighted_decay) OVER (PARTITION BY account_id, domain, month_start)
FROM weighted_decay
ORDER BY month_start
```

Platform Usage Score: Decay Score SQL

Query computes decay scores using rolling 6-month window for every account-feature pair.

```
WITH decay_score AS (  
    SELECT  
        account_id,  
        month_start,  
        domain,  
        feature_name,  
        MAX(last_usage_time) OVER (  
            PARTITION BY account_id, month_start, domain, feature_name  
            ORDER BY last_usage_time ROWS BETWEEN 6 PRECEDING AND CURRENT ROW  
        ) as max_last_usage_time,  
        coalesce((6-DATEDIFF('month', max(last_usage_time), month_start))/6, 0)  
        as decay_score  
    FROM platform_usage_score)
```

Platform Usage Score Monthly Trend



Decay And Last Usage

Year of Mo..	Month of M..	Domain	Feature Name	date_last_..	Decay S..	Weight..
2025	February	bestbuy.com	A/B/n Testing	2025-01-03	0.833	1.667
			Automated Personalizati..	2025-01-01	0.833	1.667
			Dynamic Testing	2024-12-28	0.667	1.333
			Location Targeting	Null	0.000	0.000
			Product Finder	Null	0.000	0.000
			Recommendation	Null	0.000	0.000
			Social Proofing	Null	0.000	0.000
		ebay.com	Badging	2025-01-05	0.833	0.833
			Dynamic Testing	2025-01-06	0.833	1.667
			Location Targeting	2024-12-29	0.667	0.667
			Product Finder	2025-01-01	0.833	1.667
			Recommendation	Null	0.000	0.000
		newegg.com	A/B/n Testing	Null	0.000	0.000
			Automated Personalizati..	2024-12-31	0.667	1.333
			Badging	2025-01-03	0.833	0.833
			Behavior Targeting	Null	0.000	0.000
			Cross Device Targeting	Null	0.000	0.000
			Dynamic Testing	Null	0.000	0.000
			Location Targeting	Null	0.000	0.000
			Product Finder	2024-12-31	0.667	1.333
			Social Proofing	2024-12-30	0.667	1.333
	March	bestbuy.com	A/B/n Testing	2025-02-01	0.833	1.667
			Badging	Null	0.000	0.000

- Domain
- ☐

(All)
- ☐

amazon.com
- ☒

bestbuy.com
- ☒

ebay.com
- ☐

etsy.com
- ☒

newegg.com
- ☐

overstock.com
- ☐

target.com
- ☐

walmart.com
- ☐

wayfair.com
- ☐

zappos.com

Why choose this strategy?

Prioritizing Fresh Engagement
Recent feature usage (this month) earns a high score (1.0), indicating active engagement

Identifying Trends & Gaps

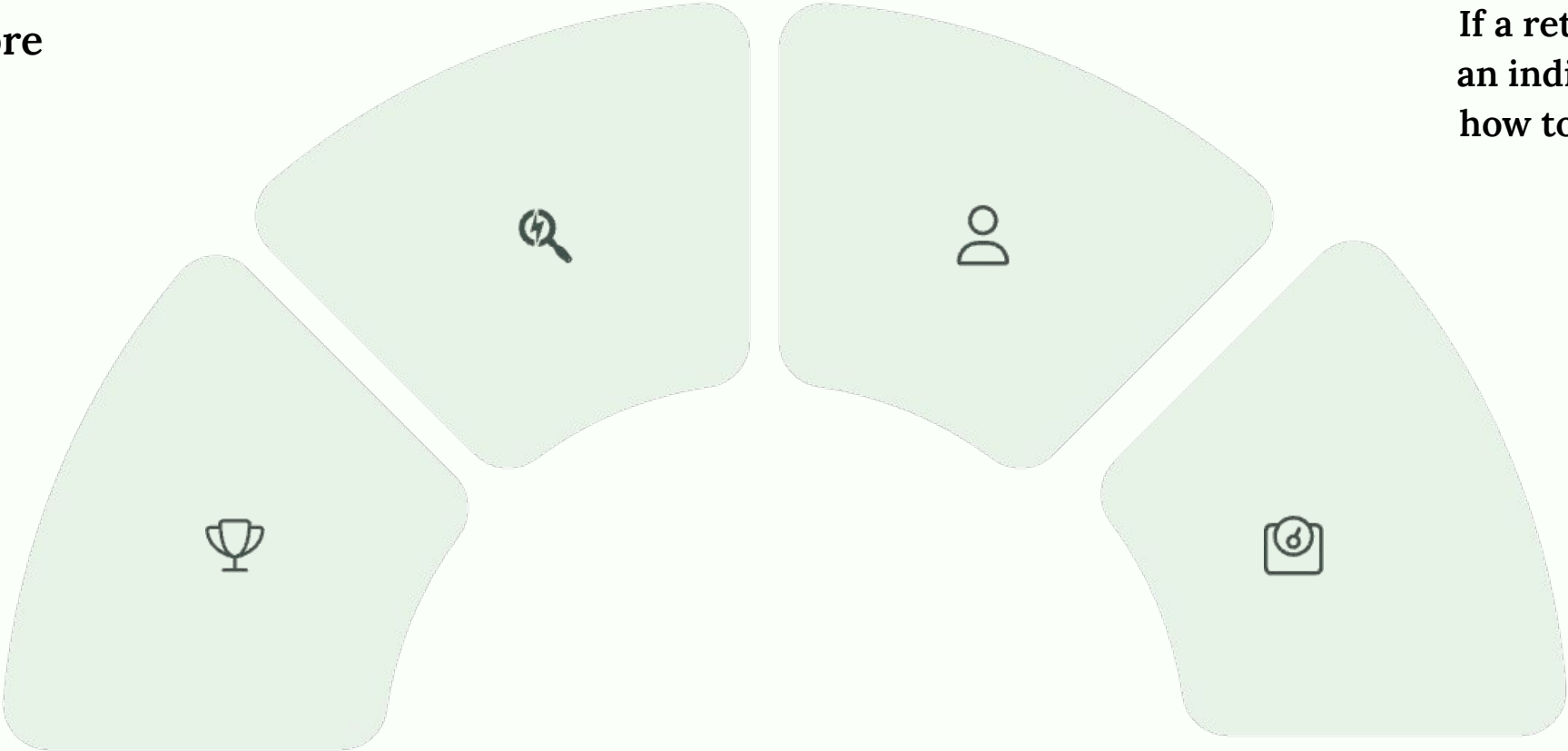
High decay scores indicate consistent feature usage and value delivery, while low scores signal opportunities for re-engagement with formerly active users

Encouraging Client Activity

Declining scores serve as early warning indicators, allowing proactive outreach for increasing user activity

Increased Awareness

If a retailer's total score is dropping, it's an indicator that they need guidance on how to make better use of the platform.



Validating the Platform Usage Score



Identify At-Risk Clients

Located clients with low scores who were approaching renewal



Conduct Targeted Meetings

Engaged with these clients to discuss usage trends



Encourage Feature Adoption

Helped clients launch more experiences



Measure Impact

Observed the impact by monitoring the retention rate, churn rate and renewal rate for 3 months time period.

Key Benefits: Retention and Revenue Impact

20%

Retention Rate Increase

Overall client retention improved after implementing targeted interventions based on Platform Usage Score insights

2

Key Client Segments Identified

Discovered clients eligible for features they weren't using and those using features after subscription expiration

Proactive Outreach

Enabled customer success teams to contact at-risk clients before renewal decisions

Glossary

Additional metrics considered before formulating platform usage score

- Churn Rate
- Customer Lifetime Value

These metrics gave information about the event that has happened. Eg. Churn rate gives information of clients who stopped using monetate. These were not much effective because they did not give us trends about usage.

Future Enhancements discussed

As per the feedback from leadership team,

- Cross feature usage:

We could observe trends for cross feature usage. Such as, what is the retention rate for people who use Dynamic testing with Auto personalization?, etc.

- Sector based scores:

We also could implement these scores based on the sector information of the clients, which needed more information and grouping of the entire client base into sectors and see how the various sectors are performing.